

SOLE SOLUTIONS

Year 8

Mid Term 1 Test

2020



Baldivis
Secondary College

Name: _____

Class: _____

Time: 50 minutes

Total Mark: ____/52

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning will be allocated zero marks.

Resources: 1 x A4 page of notes (1 sided)

Non Calculator

50 mins

Question 1

2 marks

Circle the largest number and underline the lowest number

a) -6 -4 9 -12 2 ✓

b) 0 15 -2 -21 19 ✓

Question 2

1, 1, 2, - 4 marks

Solve (showing your working)

1. $2 + 7 \times -4 = 2 - 28$
 $= -26$ ✓

2. $(5 - 3 + 2)^2 + 4 \times -5 =$
 $= 4^2 - 20$ ✓
 $= 16 - 20$
 $= -4$

3. $24 \div -2 \div -3 + 4 \times -2 - 5 =$
 $= -12 \div -3 - 8 - 5$ ✓
 $= 4 - 8 - 5$ ✓
 $= -9$ ✓

Question 3

6 marks

Evaluate:

a) $2 + 7 = 9$ ✓

b) $9 + (-3) = 6$ ✓

c) $(-17) - 3 = -20$ ✓

d) $2 - (-10) = 12$ ✓

e) $-12 - (-11) = -1$ ✓

f) $(-6) + 4 = -2$ ✓

12

Question 4**8 Marks****Evaluate:**

a) $-5 \times (-6) = 30$ ✓

b) $8 \times (-2) = -16$ ✓

c) $5 \times 7 = 35$ ✓

d) $-10 \times 3 = -30$ ✓

e) $-24 \div 3 = -8$ ✓

f) $50 \div (-5) = -10$ ✓

g) $5 + 2 \times (-3) = -1$ ✓

h) $2 \times (-7) - (-12) \div (-4) = -17$ ✓

(8)

Question 5**2 marks****Insert a $\times$ sign or $\div$ sign to make these equations true**

a) $-2 \times 3 \div (-6) = 1$ ✓

b) $-32 \times (-3) \div (-2) = -48$ ✓

2

Question 6**2 marks**

The minimum temperature in the city of Destiny was -12°C at 6am. It increased by 5°C in the next two hours and then increased by 9°C in the next five hours to reach its maximum. At what time did it reach its maximum temperature and what was this maximum temperature?

$$-12 + 5 + 9 = 2^{\circ}\text{C}$$
 ✓

$$6 + 2 + 5 = 13 \rightarrow 1 \text{ pm}$$
 ✓

2

Question 7**4 marks****For the expression $6xy + 2x - 4y^2 + 3$, state:**a) the coefficient of x 2 ✓

b) the constant term 3 ✓

c) the number of terms 4 ✓

d) the coefficient of xy 6 ✓

4

Question 8**4 marks****State whether each of the following is true or false:**a) The constant term in the expression $5x + 7$ is 5. F ✓b) $16xy$ and $5yx$ are like terms. T ✓c) The coefficient of d in the expression $6d^2 + 7d + 8abd + 3$ is 7. T ✓d) The coefficient of x and y in the expression $2x + 2y$ is 2. T ✓

4

(20)

Question 9**4 marks**Circle the *like terms* in each group

a) $2p, 3k, p$ ✓

b) $9j, 3z, -2z$ ✓

c) $-6w, 4q, -2w$ ✓

d) $14r^2, 14r, 2r^2$ ✓

4

Question 10**6 marks**

Simplify the following.

a) $4a + 3a = 7a$ ✓

b) $12d - 15d = -3d$ ✓

c) $14c + 6p + 3c = 17c + 6p$ ✓

d) $2gh + 4hg = 6gh$ ✓

e) $16g - 9g + 4g = 11g$ ✓

f) $-4h + 6h^2 + 3h + 2h^2 = 8h^2 - h$ ✓

6

Question 11**6 marks**

Simplify the following.

a) $a \times b \times b \times a \times 3 = 3a^2b^2$ ✓

b) $10a \times 4b = 40ab$ ✓

c) $4a \times bc \times 7b = 28ab^2c$ ✓

d) $2pq \times 6 \times r = 12pqr$ ✓

e) $3xy \times 8y = 24xy^2$ ✓

f) $6a \times 3a^2 = 18a^3$ ✓

6

Question 12**4 marks**

Simplify the following.

a) $10b \div b = 10$ ✓

b) $35p \div 7q = 5$ ✓

c) $\frac{25ef}{5e} = 5f$ ✓

d) $\frac{20x^2y}{36xz} = \frac{5xy}{9z}$ ✓

4

20